Kyle Willett

Senior full-stack **product engineer** with 15+ years designing, building and shipping web products, from UX to distributed backend services.

Nashville, TN
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EXPERIENCE

Independent Projects, Nashville, TN — *Product Engineer, AI-Assisted Development · kylewillett.com*

2025 - PRESENT

Design, build and ship web products with Claude Code as my implementation partner. I own the product and the engineering: specs, milestone plans, architecture decisions, code review and verification. Projects are deployed to AWS (S3 + CloudFront) with CI and analytics.

- Built a repeatable AI development workflow: written specs and decision logs, milestone plans broken into tasks, and custom Claude Code skills that plan, implement, test, commit and watch CI for each task.
- **Log Triage** (log-triage.kylewillett.com): LLM agent on the Anthropic SDK that investigates production logs through tool calls and returns ranked root causes with cited log lines, measured by an eval harness on a held-out test set. Prompt iterations raised evidence precision from 63% to ~90%, and a model comparison found the smallest model (Haiku) matching Opus at 1/15 the cost.
- **Rate Limiter** (rate-limiter.kylewillett.com): Express middleware with five rate-limiting algorithms sharing limits across instances through Redis and Lua, with in-memory failover, Prometheus metrics, a Grafana dashboard and load-balanced benchmarks.
- **The Tabernacle** (tabernacle.kylewillett.com): walkable, nighttime 3D reconstruction in Three.js with procedural geometry, custom GLSL fire and fog, positional audio and a mobile quality tier.
- **System Design Slides**: animated whiteboard walkthroughs of ten "Design X" interview questions (Stripe, Slack, Uber, Google Docs…), built in React and TypeScript.
- **Business Diagrammed**: YouTube channel explaining business and economics with narrated, code-built animated diagrams. I built the animation pipeline in TypeScript (Canvas 2D + WebGL2, rendered to ProRes by frame capture), designed the brand and chart system, and validated topics against YouTube search demand/competition.
- **Drift**: generative ambient-music canvas in React, TypeScript and Tone.js; objects on a sky play in key on a sample-accurate clock.

COMPETENCIES

Java
Scala
Typescript
Javascript
React
Node.js
HTML/CSS
SQL
NoSQL
AWS
AI / LLM / Claude Code
Kafka
Docker + Kubernetes
Akka
Postgres
MongoDB
DynamoDB
Agile/Scrum
CI/CD
Domain Driven Design
Git/GitHub
Unit Testing
UX Design

CERTIFICATES

Scrum Master (2017)

LifeWay, Nashville, TN — *Software Engineer/Senior SWE*

2009 - 2023

As an Agile/Scrum team member:

- ➔ Built enterprise B2B ecommerce application with Java and Spring, responsible for $80M in annual revenue
- ➔ Built resilient event-driven enterprise services using Scala with a focus on clustered actor systems for areas including e-commerce, online learning, and digital licensing/access, serving 1M+ users with millions of records.
- ➔ Built REST web services using both Javascript and Scala. Javascript with Node, Express, and AWS Lambda as well as Scala with the Play Framework.
- ➔ Built modern front-end javascript web applications with React, and earlier, AngularJS, utilizing EventSource for async message-driven communication.
- ➔ Utilized Kafka for flexible streamed/event-driven collaboration between micro-services.
- ➔ Collaborated with product owners and business stakeholders to gather requirements and design systems using domain driven design and event-storming.
- ➔ Delivered incremental business value utilizing tools and best practices like scrum methodology, versioning, code review, CI/CD, unit testing, end-to-end testing, and monitoring/analytics.
- ➔ Designed UX for web applications including commerce (B2B and B2C), digital curriculum, internal dashboards, and admin tools.

Jewelry Television, Nashville, TN — *Web Designer/Developer*

2007 - 2009

Initially hired as Web Designer for ShopatHome.com, then JTVwatches.com, and responsibilities expanded to Head Designer/Developer role for the Website. Created support materials for Website and marketing campaigns. Became sole designer for company, following Web Director's departure. Worked in a team environment with marketing, photographers, and more.

Kyle W Design, Midland, MI — *Web Designer/Developer and Graphic Designer*

2002 - 2007

Managed and coordinated graphic and Web design projects from initial client consultation through concept development, design and project completion. Worked with clients to determine objective, create vision, conceive designs, content and structure, consistently meeting launch deadlines within budget requirements. Used HTML, CSS, Photoshop, Illustrator, Text Editor, PHP, MySQL in Web and graphic design projects.

Midland Free, Midland, MI — *Webmaster*

2002 - 2006

Organized and edited Website content. Produced designs for marketing materials (direct mailers, booklets, large banners, bulletins). Used knowledge of color, images and effects to enhance desired message. Utilized Photoshop, Illustrator, HTML, CSS, Javascript, PHP and MySQL to complete 200-page Website and other multimedia and video projects.

EDUCATION

Cedarville University, Cedarville, OH — *Bachelor of Arts in Multimedia Technologies*

1997 -2001